

Advanced Java Lab

Week-3

Roll Number: 238W1A12A7

1. Implement batch update using JDBC.

Code:

```
import java.sql.*;

public class Mavenproject2 {

    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/mydb";
        String user = "root";
        String password = "";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);

            con.setAutoCommit(false); // REQUIRED FOR BATCH

            String sql = "INSERT INTO Person (PersonID, FirstName, City) VALUES (?, ?, ?)";

            PreparedStatement ps = con.prepareStatement(sql);

            // Record 1
            ps.setInt(1, 201);
            ps.setString(2, "Rohit");
            ps.setString(3, "Nagpur");
```

```
        ps.addBatch();

        // Record 2
        ps.setInt(1, 202);
        ps.setString(2, "Sneha");
        ps.setString(3, "Kolkata");
        ps.addBatch();

        // Record 3
        ps.setInt(1, 203);
        ps.setString(2, "Kavya");
        ps.setString(3, "Goa");
        ps.addBatch();

        // Execute all queries in one go
        int[] results = ps.executeBatch();

        con.commit();
        con.close();

        System.out.println("Batch Update Completed!");
        System.out.println("Rows affected: " + results.length);

    } catch (Exception e) {
        System.out.println("Error: " + e);
    }
}
}
```

Output:

```
--- exec:3.1.0:exec (default-cli) @ mavenproject2 ---
Batch Update Completed!
Rows affected: 3
-----
BUILD SUCCESS
-----
Total time: 4.982 s
Finished at: 2025-11-22T17:49:25+05:30
-----
```

2. Create a CRUD application using JDBC with transaction

Code:

```
package com.mycompany.mavenproject2;

import java.sql.*;

public class Mavenproject2 {
    public static void main(String[] args) {

        String url = "jdbc:mysql://localhost:3306/mydb";
        String user = "root";
        String password = "";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);

            con.setAutoCommit(false); // IMPORTANT

            // INSERT -----
```

```
String insertSQL = "INSERT INTO Person (PersonID, FirstName, City) VALUES  
(?, ?, ?)";
```

```
PreparedStatement insert = con.prepareStatement(insertSQL);
```

```
insert.setInt(1, 301);
```

```
insert.setString(2, "Anita");
```

```
insert.setString(3, "Surat");
```

```
insert.executeUpdate();
```

```
// UPDATE -----
```

```
String updateSQL = "UPDATE Person SET City=? WHERE PersonID=?";
```

```
PreparedStatement update = con.prepareStatement(updateSQL);
```

```
update.setString(1, "Vadodara");
```

```
update.setInt(2, 301);
```

```
update.executeUpdate();
```

```
// DELETE -----
```

```
String deleteSQL = "DELETE FROM Person WHERE PersonID=?";
```

```
PreparedStatement delete = con.prepareStatement(deleteSQL);
```

```
delete.setInt(1, 202); // delete existing user
```

```
delete.executeUpdate();
```

```
// If everything successful → COMMIT
```

```
con.commit();
```

```
System.out.println("Transaction Successful!");
```

```
con.close();
```

```
}
```

```
catch (SQLException e) {
```

```
    System.out.println("Database Error: " + e.getMessage());
```

```

    try {
        // ROLLBACK on failure

        Connection con = DriverManager.getConnection(url, user, password);

        con.rollback();

        System.out.println("Transaction Rolled Back!");
    } catch (SQLException ex) {

        System.out.println("Rollback Failed: " + ex.getMessage());

    }
}

catch (Exception ex) {

    System.out.println("Error: " + ex.getMessage());

}

}

}

```

Output:

```

--- exec:3.1.0:exec (default-cli) @ mavenproject2 ---
Transaction Successful!
-----
BUILD SUCCESS
-----
Total time:  5.410 s
Finished at: 2025-11-22T17:51:47+05:30
-----

```